

Predictive Fault Detection Report

Clean summary of the run, metrics, and generated artifacts

Generated 02 Apr 2026, 17:25

Dataset	predictive_maintenance.csv	Input Shape	(10, 6)	Classes	6
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A polished summary of the run, metrics, and generated artifacts.

Run Overview

Field	Value
Generated	2026-04-02 17:25:34
Dataset	predictive_maintenance.csv
Input Shape	(10, 6)
Classes	Heat Dissipation Failure, No Failure, Overstrain Failure, Power Failure, Random Failures, Tool Wear Failure

Run Settings

Setting	Value
Epochs	1
Batch	64
Timesteps	10
Simulator	Disabled
Simulation Steps	1

Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Binary LSTM	0.9804	0.0000	0.0000	0.0000	0.8549
Multiclass LSTM	0.9794	0.9631	0.9794	0.9712	N/A

Binary Accuracy	0.980	Multiclass Accuracy	0.979	Elapsed	0.4 min
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Generated Files

Status	File	Description
OK	binary_training_history.png	Binary LSTM loss & accuracy curves
OK	binary_cm.png	Binary confusion matrix
OK	binary_roc.png	Binary ROC curve

OK	binary_f1.png	Binary F1 bar chart
OK	multiclass_training_history.png	Multiclass training curves
OK	multiclass_cm.png	Multiclass confusion matrix
OK	multiclass_roc.png	Multiclass ROC curves (one-vs-rest)
OK	multiclass_f1.png	Multiclass F1 bar chart
OK	results_summary.csv	Combined metrics table
SKIPPED	simulation_log.csv	Robot simulator step-by-step log
OK	Report.pdf	Clean PDF report

Elapsed time: 0.4 min